BERNSTEIN
SOCIETE GENERALE GROUP

Weekend Healthcare Pulse: The heavy side of weight loss - peptide manufacturing, waste, and ways forward

Rebecca Liang, Ph.D. +852 2123 2656 rebecca.liang@bernsteinsg.com
Miki Sogi, Ph.D. +81 3 6777 6991 miki.sogi@bernsteinsg.com
Courtney Breen +1 917 344 8407 courtney.breen@bernsteinsg.com
Justin Smith +44 20 7762 5899 justin.smith@bernsteinsg.com
Eve Burstein +1 917 344 8313 eve.burstein@bernsteinsg.com
Lee Hambright +1 917 344 8429 lee.hambright@bernsteinsg.com
Susannah Ludwig +41 582 723 127 susannah.ludwig@bernsteinsg.com
William Pickering, MD +1 917 344 8340 william.pickering@bernsteinsg.com
Lance Wilkes +1 917 344 8501 lance.wilkes@bernsteinsg.com
Delphine Le Louet +33 1 42 13 92 93 delphine.le-louet@bernsteinsg.com

Related reading - Weekend Healthcare Pulse: GLP-1 peptides, from bench to bedside - a manufacturing view

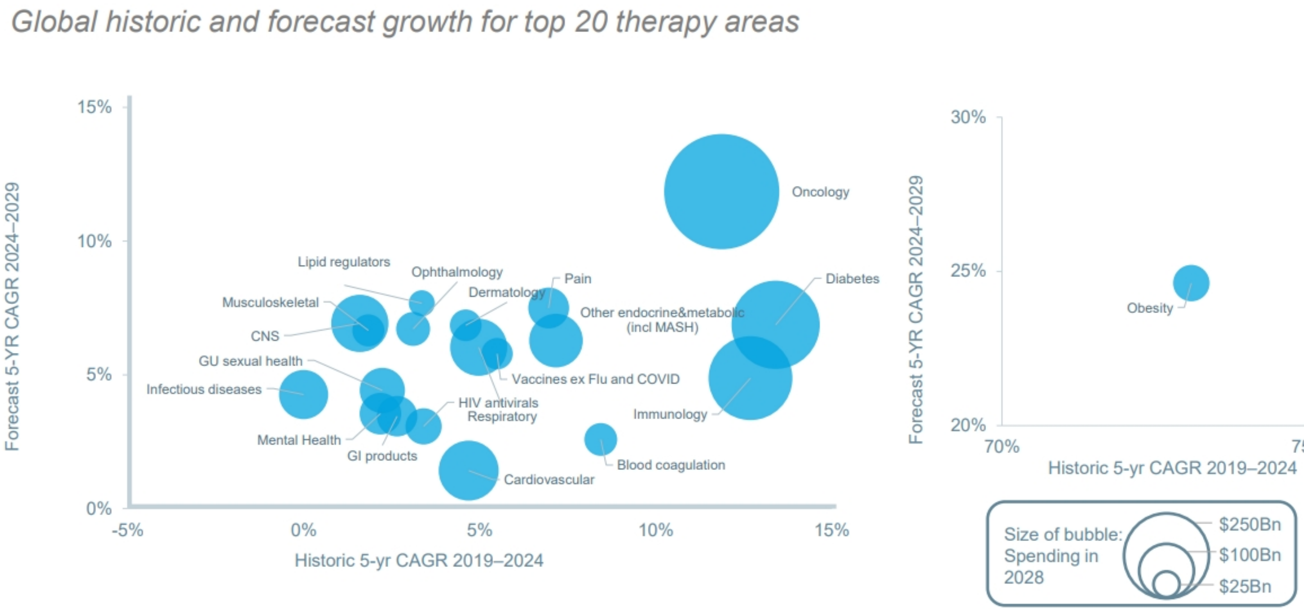
PEPTIDES - LARGEST MARKET AMONG NEW DRUG MODALITIES

By 2030, IQVIA projects medicine use roughly 4 trillion defined daily doses globally. Across the top 20 therapy areas, most see mid-single to low-teens growth, but **obesity is in its own quadrant and forecast to grow fastest over the next five years**, well ahead of diabetes, oncology and immunology. Global obesity drug spending is projected to compound in the mid-20s, adding more than USD 50 Bn of incremental spend between 2025 and 2029 and pushing the category **toward the USD 100 Bn threshold** by the end of the decade.

Growth in the obesity market pushes peptide forward as the top new drug modality in size and growth, beating ADCs, next-generation antibodies and oligonucleotides. While recombinant proteins and conventional antibodies still dominate in absolute value, the growth bubble for peptides is now comparable to those established biologic classes.

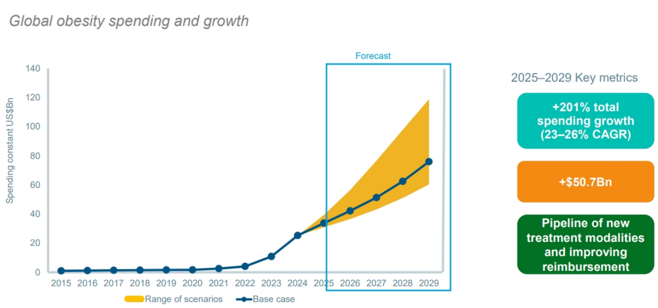
First Published: 05 Jun 2026 09:25 UTC Completion Date: 05 Jun 2026 09:25 UTC

EXHIBIT 1: **Oncology and obesity are the fastest growing therapeutic areas - and should remain so in the near term**



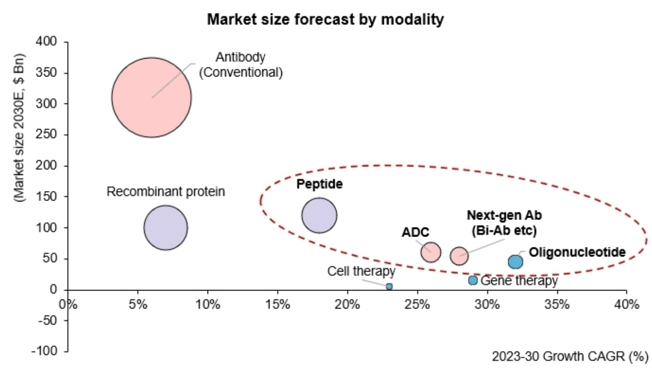
Source: IQVIA "The Global Use of Medicines 2025: Outlook to 2029"

EXHIBIT 2: **Depending on pricing and reimbursement developments, obesity spending could break into the $100Bn level in the coming years**



Source: IQVIA "The Global Use of Medicines 2025: Outlook to 2029"

EXHIBIT 3: **Among the newer molecular modalities, peptides stand out as both the largest market by 2030 and one of the fastest-growing**



Source: Fujifilm, Bernstein analysis

Top peptide drugs drive a global manufacturing capacity crunch. Flagship GLP-1 products like semaglutide, tirzepatide, and liraglutide each rely on long, chemically synthesized peptides or hybrid recombinant-chemical routes. And GLP-1s are not the only driver: blockbusters such as octreotide, leuprolide, goserelin, teriparatide and lanreotide span endocrinology, oncology, osteoporosis and neuroendocrine tumors, each with its own manufacturing route (pure SPPS, hybrid, or recombinant DNA). The combined demand boom catapults the supply side - peptide synthesis - from a niche segment reserved for academic research and cosmetics a decade ago into a major industry of pharmaceutical manufacturing.

EXHIBIT 4: **Representative peptide-based drugs and their manufacturing**

Brand Name	Generic Name	Drug Target	Therapeutic Indications	Primary Manufacturing Method
Ozempic / Wegovy	Semaglutide	GLP-1R	Type 2 Diabetes, Obesity / Weight management	**Hybrid** (Recombinant backbone + Chemical modification)
Mounjaro / Zepbound	Tirzepatide	GIPR / GLP-1R (Dual agonist)	Type 2 Diabetes, Obesity / Weight management	**Chemical** (SPPS / LPPS Hybrid)
Victoza / Saxenda	Liraglutide	GLP-1R	Type 2 Diabetes, Obesity / Weight management	**Hybrid** (Recombinant backbone + Chemical modification)
Copaxone	Glatiramer acetate	MHC Class II molecules (Immunomodulator)	Relapsing Multiple Sclerosis (MS)	**Chemical** (Solution/Solid-phase polymerization)
Trulicity	Dulaglutide	GLP-1R	Type 2 Diabetes	**Recombinant DNA** (Mammalian cell culture)
Sandostatin	Octreotide	SSTR (Somatostatin receptors, primarily SSTR2/5)	Acromegaly, Severe diarrhea / Carcinoid syndrome	**Chemical** (SPPS)
Lupron / Eligard	Leuprolide	GnRHR (Gonadotropin-releasing hormone receptor agonist)	Prostate Cancer, Endometriosis, Central Precocious Puberty	**Chemical** (SPPS)
Forsteo / Forteo	Teriparatide	PTH1R (Parathyroid hormone receptor type 1)	Severe Osteoporosis	**Recombinant DNA** (*E. coli* expression)
Zoladex	Goserelin	GnRHR	Prostate Cancer, Breast Cancer, Endometriosis	**Chemical** (SPPS)
Somatuline	Lanreotide	SSTR	Acromegaly, Neuroendocrine Tumors (NETs)	**Chemical** (SPPS)

Source: company disclosure, Bernstein analysis

FROM KG TO MULTI-METRIC TON MANUFACTURING

Historically, peptide APIs were produced in kilograms per year, and now they're entering multi-kilogram and even metric-ton annual volumes. Most marketed peptide APIs are manufactured either by bioreactors (recombinant expression followed by purification) or by chemical synthesis, with solid-phase peptide synthesis (SPPS) the dominant route for short- to medium-length and heavily modified peptides. SPPS builds the peptide stepwise on a resin with iterative deprotection, coupling, and washing cycles, then cleaves and purifies the crude sequence by chromatography. The process is flexible (e.g. can incorporate non-natural amino acids, lipidation) and compatible with automation, which is why it has become the industry standard for modern peptide APIs despite its heavy solvent usage.

SPPS and batch chromatography (for purification) are extremely solvent- and reagent-consuming: industry analyses suggest that up to **16,000 kg of chemicals are needed per kg of GLP-class peptide API**, with purification and lyophilization major contributors to process mass intensity (PMI). At larger scales, any inefficiency in coupling, deprotection, or chromatography multiplies into higher COGS and more hazardous waste. If the API is of long sequences and hydrophobic segments, this would increase aggregation and side-product formation, making high-purity specs harder to hit. **ESG is, consequentially, a problem: organizations like the ACS Green Chemistry Institute explicitly flag peptide APIs as among the least sustainable modalities.**

There three major challenges peptide manufacturing scales:

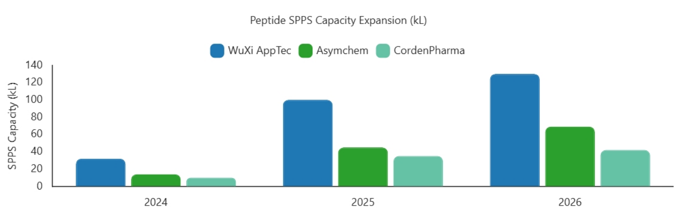
- High PMI and solvent consumption **threaten margin sustainability**: if CDMOs cannot meaningfully cut solvent and raw-material use per kg, rising volume magnifies costs and environmental liabilities.
- Batch purification and complex multi-step workflows limit effective capacity and raise the **probability of batch failures**, constraining revenue growth and creating supply risk overhangs.
- **ESG and regulatory expectations** around waste minimization and impurity control are tightening; players that remain on legacy solvent-heavy platforms risk losing share, facing higher capex, or less sticky contracts with customers that prefer greener synthesis routes.

EXHIBIT 5: **Comparison of peptide synthesis methods - despite high solvent waste, SPPS is the commercial standard for mid to long peptides like GLP-1 agonists**

Synthesis Method	Target Molecules	Technological differences	Commercial adoption	CDMO Execution
SPPS	• Median length peptides (20–45+ AAs)	• Heterogeneous phase reactions • High solvent/reagent footprint	**Commercial standard for GLP-1** and other other peptide API > 10 aa despite high waste	**Bachem**: Multi-metric ton scale via automated arrays in Building K. **WuXi AppTec**: Operating over 100,000L of SPPS capacity.
LPPS	• Ultra-short peptides or fragments (<10 AAs) • Cosmetic peptides	• Homogeneous solution kinetics Low solvent waste Requires manual workup for every AA addition	Rarely used for long GLP-1s; sequential extractions cause massive yield loss and delay timelines.	**EuroAPI**: Partnered with StrainChem for eco-friendly, short-chain production.
Hybrid (SPPS+LPPS)	•Macrocyclic/branched peptides • Very long chains (>40–50 AAs)	• Fragments made via SPPS and linked by LPPS • Risk of racemization at link sites	Used for long/complex peptides and increases crude purity	**PolyPeptide**: Links SPPS fragments in solution at its Belgium site.
TAPS (Tag-assisted peptide synthesis; liquid phase)	• Short to mid length or difficult sequences	• Soluble molecular anchor tag replaces resin in SPPS • Reduces solvent waste	Relatively early in adoption curve and reserved for high-value API	**CordenPharma**: Scaled TAPS via AmbioPharm to run in standard chemical reactors and achieve hundred-kg scale

Source: ACS, C&EN, Bernstein analysis

EXHIBIT 6: **Major CDMOs are expanding their peptide synthesis capacity to meet demand growth**



Source: C&EN, company disclosure, Bernstein analysis

EXHIBIT 7: **Peptide synthesis is a dirty process that consumes high amounts of hazardous solvents like N,N-dimethylformamide (DMF), N-methyl-2-pyrrolidone, and dichloromethane**

BY THE NUMBERS

14 t

Amount of solvent that can be consumed to produce 1 kg of GLP-1 agonist peptide

0.3 t

Amount of solvent that can be consumed to produce 1 kg of a typical small-molecule drug

Source: https://cen.acs.org/pharmaceuticals/pharmaceutical-chemicals/making-weight-loss-drugs-mean/102/i13

INNOVATIVE IMPROVEMENTS ON SPPS - CASE STUDIES ON CDMOS

Bachem - pioneer of continuous chromatography (MCSGP)

MCSGP (multi-column countercurrent solvent gradient purification) was originally developed by researchers at ETH Zürich. **Bachem (not covered) was the first CDMO to commercialize the technology in 2023-24**. Instead of running big, stop-start purification steps and throwing away a lot of in-between material, MCSGP uses two connected columns that continuously pass partially pure fractions back and forth until the peptide is cleanly separated. In practice, this means a much larger share of what used to be waste or rework can be recovered as saleable product, while still hitting very high purity of over 99.3%.

Compared to traditional batch purification, **MCSGP cuts solvent consumption by about 30% and total material use by 70%**. At the same time, what used to take 5 days of batch work can be squeezed into a 24-hour continuous run. That cuts in-process control checks, shrinks lyophilizate volumes, and shortens purification from about five days to one, freeing up both equipment and QC resources.

EXHIBIT 8: **Compared to traditional batch chromatography, continous chromatography (multi-column countercurrent solvent gradient purification or MCSGP) increases throughput and reduces process control and lyphilization volume**

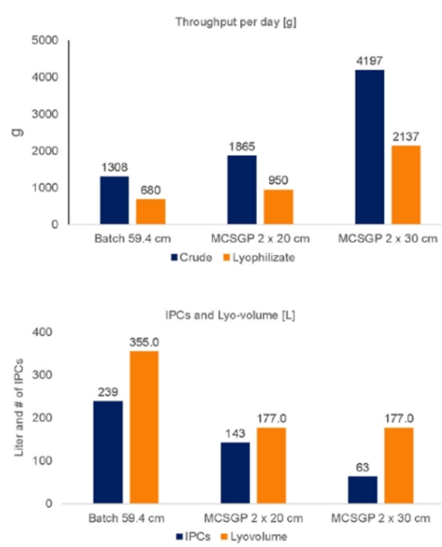


Figure 2 : Comparison of throughput, IPCs and lyophilizate volumes between batch and MCSGP processes

Source: Bachem

WuXi Apptec - reducing solvent use and achieving over 3X lower PMI

In optimizing their process, Wuxi Apptec (not covered) goes squarely after solvent reduction. **They have cut total solvent use by 25% and replaced 50% of the DMF with more sustainable solvents.** The remainder DMF is recycled and used by nearby battery plants to minimize overall waste of the industrial complex.

PMI is the total mass of all input materials required to produce one unit mass of product; **a lower PMI directly reflects a less resource-intensive process**. Industry analyses of current peptide manufacturing report average PMIs in the 13,000 range for solid-phase peptide synthesis. WuXi's latest data show they've cut that by a factor of over 3X. Fewer solvents and reagents lower variable costs and waste-handling spend. Shaving 2/3 off the material footprint per kilo adds up quickly in terms of margin. Besides, it frees up capacity. Last but not least - pharma companies are under pressure to report Scope 3 emissions; if they outsource to a CDMO that can show quantified PMI reductions and solvent recycling, that makes their sustainability narrative easier. That tends to translate into stickier contracts. **Lower PMI supports better unit economics, better capital productivity, and a differentiated ESG offering.**

EXHIBIT 9: **In 2025, Wuxi Apptec announced optimization of PMI relative to industry benchmark by over 3X - reducing solvent and waste in peptide synthesis**

Table 1. Measurable Impact on PMI: WuXi TIDES vs. Industry Benchmark

	Peptide PMI - Industry Benchmark[1]	Peptide PMI - WuXi TIDES[2]	Fold Improvement
R&D	33,141	9,399	**3.5x**
Commercial Manufacturing	13,063	3,659	**3.6x**

[1]Source: Kekessie I *et al. J Org Chem*. 2024;89(7):4261-4282.
[2] WuXi TIDES data based on projects completed in 2024

Source: https://tides.wuxiapptec.com/resources/lower-pmi-in-peptide-synthesis/

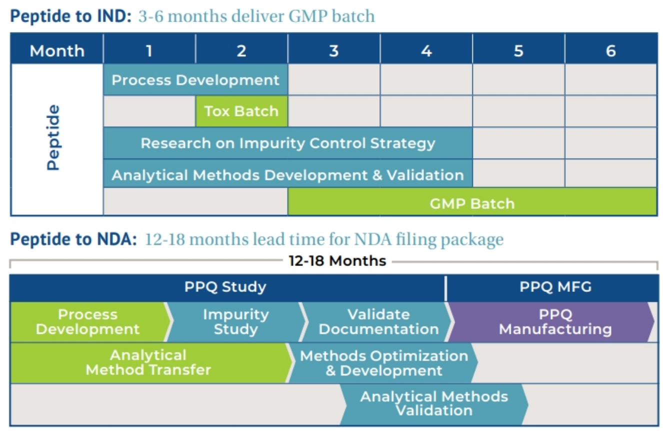
Asymchem - Delivering PPQ within a year

We see similar methods adopted by Asymchem (not covered):

- Greener synthetic routes (DMF substitution and sustainable solvents)
- Real-time Process Analytical Technology (PAT) and predictive kinetic modeling to reduce wash steps and experiments, and
- Tag-assisted peptide synthesis (TAPS) plus continuous flow reactors to enhance mixing, reaction rates, and scalability

Typical PPQ (Process Performance Qualification) programs for complex long peptides and GLP-1s tend to run on the order of 18–24 months. Asymchem's Mazdutide case study—completing pre-PPQ and PPQ in 12 months, submitting the NDA dossier in 6 months, passing a dynamic PAI within 6 months, and delivering a >5 kg, >98.5% purity batch. This looks roughly 6–12 months faster than what we would expect for a first-in-class dual GCG/GLP-1 agonist at ton-scale, while still meeting high purity and impurity-control expectations.

EXHIBIT 10: **Asymchem case study: 3-6m to deliver an IND project and 12-18m to deliver an NDA project**



Source: Asymchem

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	5 Jun 2026 Closing Price	Price Target	TTM Rel. Perf.	Reported EPS Cur	Reported EPS 2025A	Reported EPS 2026E	Reported EPS 2027E	Reported P/E (x) 2025A	Reported P/E (x) 2026E	Reported P/E (x) 2027E
9926.HK (Akeso)	M	HKD	93.80	130.00	(31.2)%	CNY	(0.60)	(0.52)	0.70	(134.1)	(157.1)	116.1
ONC (BeOne)	O	USD	279.93	412.00	(18.1)%	USD	2.63	5.65	8.93	106.4	49.5	31.3
1093.HK (CSPC)	M	HKD	7.26	10.70	(53.3)%	HKD	0.37	0.52	0.56	17.0	12.2	11.2
3692.HK (Hansoh)	O	HKD	29.26	47.00	(35.0)%	CNY	0.93	1.01	1.10	27.2	25.0	23.0
1801.HK (Innovent)	O	HKD	76.60	120.00	(39.7)%	HKD	0.48	0.71	2.90	157.9	107.9	26.4
600276.CH (Hengrui)	O	CNY	47.08	71.00	(58.3)%	CNY	1.19	1.54	1.92	39.6	30.5	24.5
6990.HK (Kelun-Biotech)	O	HKD	393.40	526.00	(27.1)%	CNY	(1.66)	(2.75)	0.85	32.6	33.2	20.4
1177.HK (Sino BioPh)	M	HKD	4.65	7.90	(43.0)%	CNY	0.19	0.20	0.22	21.0	20.3	18.4
9688.HK (Zai Lab)	M	HKD	12.60	15.00	(101.5)%	USD	(0.16)	(0.15)	(0.09)	0.4	0.4	0.3
MRK (Merck)	M	USD	120.26	100.00	27.9%	USD	8.97	5.06	9.21	13.4	23.8	13.1
BMY (Bristol-Myers Squibb)	M	USD	56.60	58.00	(8.7)%	USD	6.15	6.31	6.02	9.2	9.0	9.4
PFE (Pfizer)	M	USD	25.69	30.00	(15.9)%	USD	3.23	2.92	2.57	8.0	8.8	10.0
ASIAX			2,024.37									
SPX			7,584.31									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

MRK, BMY, PFE estimate is Adjusted EPS; 6990.HK, 9688.HK valuation is EV/Sales (x); MRK, BMY, PFE valuation is Adjusted P/E (x); 9926.HK, 1093.HK, 1177.HK base year is 2024;

Source: Bloomberg, Bernstein estimates and analysis.

I. REQUIRED DISCLOSURES

References to "Bernstein" or the “Firm” in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK（サンフォード・C・バーンスタイン株式会社）and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein’s “affiliates” relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY

This research publication covers six or more companies. For valuation methodology and other company disclosures:
Please visit: https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action.
Or, you can also write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

RISKS

This research publication covers six or more companies. For risks and other company disclosures:
Please visit: https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action.
Or, you can also write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

Bernstein brand

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/-15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 500 Banks And Financial Services Index (BEBANKS) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 500 Insurance Index (BEINSUR) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/-10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- Credit Outperform (C-OP): The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Neutral (C-N): The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- Credit Underperform (C-UP): The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.1%	16.5%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	36.3%	17.8%
Underperform	SELL	12.6%	14.9%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.
As of March 31, 2026. All figures are updated quarterly.

PRICE CHARTS/ RATINGS AND PRICE TARGET HISTORY

This research publication covers six or more companies. For price chart and other company disclosures, please visit https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action or you can write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

CONFLICTS OF INTEREST

Bernstein and/or affiliates have received compensation for investment banking services in the past twelve months from Merck & Co Inc.

An affiliate of Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Merck & Co Inc and Pfizer Inc.

Bernstein and/or affiliates have received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Merck & Co Inc, Bristol-Myers Squibb Co and Pfizer Inc.

Affiliates of Bernstein managed or co-managed in the past twelve months a public offering of securities of Merck & Co Inc.

Bernstein and/or affiliates had an investment banking client relationship during the past twelve months with Merck & Co Inc.

Certain affiliates of Bernstein act as market maker or liquidity provider in the equities securities of: Merck & Co Inc, Bristol-Myers Squibb Co and Pfizer Inc.

Certain affiliates of Bernstein act as market maker or liquidity provider in the debt securities of: Merck & Co Inc and Pfizer Inc.

OTHER MATTERS

The legal entity(ies) employing the analyst(s) listed in this report, and their location, can be determined by the country code of their phone number, as follows:

+1 Bernstein Institutional Services LLC; New York, New York, USA

+44 Bernstein Autonomous LLP; London UK

+212 Société Générale Africa Technologies & Services; Casablanca, Morocco

+33 BSG France S.A.; Paris, France

+34 BSG France S.A.; Madrid, Spain

+41 Bernstein Autonomous LLP; Geneva, Switzerland

+49 BSG France S.A.; Frankfurt, Germany

+91 Sanford C. Bernstein (India) Private Limited; Mumbai, India

+852 Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司; Hong Kong, China

+65 Sanford C. Bernstein (Singapore) Private Limited; Singapore

+81 Sanford C. Bernstein Japan KK; Tokyo, Japan

Where this report has been prepared by research analyst(s) employed by a non-US affiliate, such analyst(s), is/are (unless otherwise expressly noted below) not registered as associated persons of Bernstein Institutional Services LLC or any other SEC-registered broker-dealer and are not licensed or qualified as research analysts with FINRA. Accordingly, such analyst(s) may not be subject to FINRA's restrictions regarding (among other things) communications by research analysts with a subject company, interactions between research analysts and investment banking personnel, participation by research analysts in solicitation and marketing activities relating to investment banking transactions, public appearances by research analysts, and trading securities held by a research analyst account.

Where this report has been prepared by research analyst(s) employed by Société Générale Africa Technologies & Services (part of the Société Générale group of companies), it has been prepared on behalf of a Bernstein company under a Global Services Agreement in place between Bernstein and Société Générale.

CERTIFICATION

Each research analyst listed in this report, who is primarily responsible for the preparation of the content of this report, certifies that all of the views expressed in this publication accurately reflect that analyst's personal views about any and all of the subject securities or issuers and that no part of that analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views in this publication.

II. ADDITIONAL GLOBAL CONFLICT DISCLOSURES

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e., the private side) within the Firm, and into other areas, units, groups or affiliates (i.e., public side) of the Firm.

III. OTHER IMPORTANT INFORMATION AND DISCLOSURES

Separate branding is maintained for “Bernstein” and “Autonomous” research products.

- Bernstein produces a number of different types of research products including, among others, fundamental analysis and quantitative analysis under both the “Autonomous” and “Bernstein” brands. Recommendations contained within one type of research product may differ from recommendations contained within other types of research products, whether as a result of differing time horizons, methodologies or otherwise. Furthermore, views or recommendations within a research product issued under one brand may differ from views or recommendations under the same type of research product issued under the other brand. The Research Ratings System for the two brands and other information related to those Rating Systems are included in the previous section.

- Autonomous operates as a separate business unit within the following entities: Bernstein Institutional Services LLC, Bernstein Autonomous LLP, Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 and Sanford C. Bernstein (India) Private Limited. For information relating to “Autonomous” branded products (including certain Sales materials) please visit: www.autonomous.com. For information relating to Bernstein branded products please visit: www.bernsteinresearch.com.

Analysts are compensated based on aggregate contributions to the research franchise as measured by account penetration, productivity and proactivity of investment ideas. No analysts are compensated based on performance in, or contributions to, generating investment banking revenues.

This report has been produced by an independent analyst as defined in Article 3 (1)(34)(i) of EU 596/2014 Market Abuse Regulation (“MAR”) and the same article of MAR as it forms part of United Kingdom domestic law by virtue of the European Union (Withdrawal) Act 2018.

To our readers in the United States: Bernstein Institutional Services LLC, a broker-dealer registered with the U.S. Securities and Exchange Commission (“SEC”) and a member of the U.S. Financial Industry Regulatory Authority, Inc. (“FINRA”) is distributing this publication in the United States and accepts responsibility for its contents. Where this material contains an analysis of debt product(s), such material is intended only for institutional investors and is not subject to the US independence and disclosure standards applicable to debt research prepared for retail investors.

Bernstein Institutional Services LLC may act as principal for its own account or as agent for another person (including an affiliate) in sales or purchases of any security which is a subject of this report. This report does not purport to meet the objectives or needs of any specific individuals, entities or accounts.

To our readers in Canada: If this publication pertains to a Canadian domiciled company, it is being distributed in Canada by Sanford C. Bernstein (Canada) Limited, which is licensed and regulated by the Canadian Investment Regulatory Organization. If the publication pertains to a non-Canadian domiciled company, it is being distributed by Bernstein Institutional Services LLC, which is licensed and regulated by both the SEC and FINRA, into Canada under the International Dealers Exemption.

This document may not be passed onto any person in Canada unless that person qualifies as "permitted client" as defined in Section 1.1 of NI 31-103.

To our readers in Brazil: This report has been prepared by Bernstein Institutional Services LLC, and Banco BTG Pactual S.A. ("BTG") is responsible for the distribution of this report in Brazil.

To readers in the United Kingdom: This publication has been issued or approved for issue in the United Kingdom by Bernstein Autonomous LLP, authorised and regulated by the Financial Conduct Authority and located at 60 London Wall, London EC2M 5SH, +44 (0)20-7170-5000. Registered in England & Wales No OC343985.

This document is for distribution only to persons who (i) have professional experience in matters relating to investments falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the “Financial Promotion Order”), (ii) are persons falling within Article 49(2)(a) to (d) (“high net worth companies, unincorporated associations, etc.”) of the Financial Promotion Order, (iii) are outside the United Kingdom, or (iv) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the FSMA) in connection with the issue or sale of any securities may otherwise lawfully be communicated or caused to be communicated (all such persons together being referred to as “relevant persons”). This document is directed only at relevant persons and must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is available only to relevant persons and will be engaged in only with relevant persons.

To our readers in the member states of the EEA: This publication is being distributed by BSG France SA, which is authorised and regulated by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and Autorité des Marchés Financiers (AMF).

To our readers in Hong Kong: This publication is being distributed in Hong Kong by Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, which is licensed and regulated by the Hong Kong Securities and Futures Commission (Central Entity No. AXC846) to carry out Type 4 (Advising on Securities) regulated activities and subject to the licensing conditions mentioned in the SFC Public Register (https://www.sfc.hk/publicregWeb/corp/AXC846/details)). This publication is solely for professional investors, as defined in the Securities and Futures Ordinance (Cap. 571). The purpose of this report is solely to provide an analysis of the issuers referred to in this report and is not intended for any purpose contrary to the laws of Hong Kong.

To our readers in Singapore: This publication is being distributed in Singapore by Sanford C. Bernstein (Singapore) Private Limited, only to accredited investors or institutional investors, as defined in the Securities and Futures Act 2001 of Singapore ("SFA"). Recipients in Singapore should contact Sanford C. Bernstein (Singapore) Private Limited in respect of matters arising from, or in connection with, this publication. Sanford C. Bernstein (Singapore) Private Limited is regulated by the Monetary Authority of Singapore and licensed under the SFA as a capital markets services licence holder for dealing in capital markets products that are securities and collective investment schemes and an exempt financial adviser for advising on, issuing and promulgating analyses and reports on securities. Sanford C. Bernstein (Singapore) Private Limited is registered in Singapore with Company Registration No. 20213710W and located at One Raffles Quay, #27-11 South Tower, Singapore 048583, +65-6230-4612.

To our readers in the People's Republic of China: The securities referred to in this document are not being offered or sold and may not be offered or sold, directly or indirectly, in the People's Republic of China (for such purposes, not including the Hong Kong and Macau Special Administrative Regions or Taiwan, the "PRC") in contravention of any applicable laws of the PRC.

This document does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC to any person to whom it is unlawful to make the offer or solicitation in the PRC.

We do not represent that this document may be lawfully distributed, or that any securities may be lawfully offered, in compliance with any applicable registration or other requirements in the PRC, or pursuant to an exemption available thereunder, or assume any responsibility for facilitating any such distribution or offering. In particular, no action has been taken by us which would permit a public offering of any securities or distribution of this document in the PRC. Accordingly, the securities are not being offered or sold within the PRC by means of this document or any other document. Neither this document nor any advertisement or other offering material may be distributed or published in the PRC, except under circumstances that will result in compliance with any applicable laws and regulations.

To our readers in Japan: This publication is being distributed in Japan by Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社), which is registered in Japan as a Financial Instruments Business Operator with the Kanto Local Finance Bureau (registration number: The Director-General of Kanto Local Finance Bureau (FIBO) No.3387) and regulated by the Financial Services Agency. It is also a member of Japan Investment Advisers Association. This publication is solely for qualified institutional investors in Japan only, as defined in Article 2, paragraph (3), items (i) of the Financial Instruments and Exchange Act.

For the institutional client readers in Japan who have been granted access to the Bernstein website by Daiwa Securities Group Inc. ("Daiwa"), your access to this document should not be construed as meaning that Bernstein is providing you with investment advice for any purposes. Whilst Bernstein has prepared this document, your relationship is, and will remain with, Daiwa, and Bernstein has neither any contractual relationship with you nor any obligations towards you.

To our readers in Australia: Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is responsible for distributing research in Australia. It is regulated by the Securities and Exchange Commission under U.S. laws, by the Financial Conduct Authority under U.K. laws, which differs from Australian laws. Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is exempt from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the provision of the following financial services to wholesale clients:

- providing financial product advice;
- dealing in a financial product;
- making a market for a financial product; and

- providing a custodial or depository service.

To our readers in India: This publication is being distributed in India by Sanford C. Bernstein (India) Private Limited (SCB India) which is licensed and regulated by Securities and Exchange Board of India ("SEBI") as a research analyst entity under the SEBI (Research Analyst) Regulations, 2014, having registration no. INH000006378 and as a stock broker having registration no. INZ000213537. SCB India is currently engaged in the business of providing research and stock broking services. Please refer to www.bernsteinresearch.in for more information.

- SCB India is a Private limited company incorporated under the Companies Act, 2013, on April 12, 2017 bearing corporate identification number U65999MH2017FTC293762, and registered office at Level 3A, 4th Floor, First International Financial Centre, Plot Nos C-54 and C-55, G Block, Near CBI Office, Bandra Kurla Complex, Bandra (East), Mumbai 400098, Maharashtra, India (Phone No: +91-22-68421401).
- For details of Associates (i.e., affiliates/group companies) of SCB India, kindly email MUM-BERNSTEIN-InCompliance@bernsteinsg.com.
- SCB India does not have any disciplinary history as on the date of this report.
- Except as noted above, SCB India and/or its Associates (i.e., affiliates/group companies), the Research Analysts authoring this report, and their relatives
  - do not have any financial interest in the subject company
  - do not have actual/beneficial ownership of one percent or more in securities of the subject company;
  - is not engaged in any investment banking activities for Indian companies, as such;
  - have not managed or co-managed a public offering in the past twelve months for any Indian companies;
  - have not received any compensation for investment banking services or merchant banking services from the subject company in the past 12 months;
  - have not received compensation for brokerage services from the subject company in the past twelve months;
  - have not received any compensation or other benefits from the subject company or third party related to the specific recommendations or views in this report; and
  - do not currently, but may in the future, act as a market maker in the financial instruments of the companies covered in the report.
  - do not have any conflict of interest in the subject company as of the date of this report.
- Except as noted above, the subject company has not been a client of SCB India during twelve months preceding the date of distribution of this research report. Neither SCB India nor its Associates (i.e., affiliates/group companies) have received compensation for products or services other than investment banking, merchant banking or brokerage services from the subject company in the past twelve months.
- The principal research analyst(s) who prepared this report, members of the analysts' team, and members of their households are not an officer, director, employee or advisory board member of the companies covered in the report.
- Our Compliance officer / Grievance officer is Ms. Rupal Talati, who can be reached at +91-22-68421451, or MUM-BERNSTEIN-InCompliance@bernsteinsg.com / Scbin-investorgrievance@bernsteinsg.com
- The Research investor charter and Terms & Conditions of SCB India are available on its website and may be accessed at Sanford C. Bernstein (India) Private Limited (https://bernsteinresearch.in/) for your reference.
- Disclaimer: Registration granted by SEBI, and certification from NISM, is in no way a guarantee of performance of the intermediary or provide any assurance of returns to investors. Investments in securities market are subject to market risks. Read all the related documents carefully before investing.

To our readers in Switzerland: This document is provided in Switzerland by or through Bernstein Autonomous LLP, and is provided only to qualified investors as defined in article 10 of the Swiss Collective Investment Scheme Act ("CISA") and related provisions of the Collective Investment Scheme Ordinance and in strict compliance with applicable Swiss law and regulations. The products mentioned in this document may not be suitable for all types of investors. This document is based on the Directives on the Independence of Financial Research issued by the Swiss Bankers Association (SBA) in January 2008.

To our readers in the Middle East: Bernstein Autonomous LLP, DIFC branch has its principal office at Gate Village 06, DIFC, Dubai, UAE. Bernstein Autonomous LLP, DIFC branch is regulated by the Dubai Financial Services Authority (DFSA) with the registration number CL10040 and is provisioned for Arranging Deals in Investments and Advising on Financial Products. All communications and services are directed at Professional Clients and Market Counterparties only (as defined in the DFSA rulebook). Persons other than Professional Clients and Market Counterparties, such as Retail Clients, are not the intended recipients of our communications or services.

LEGAL

All research publications are disseminated to our clients through posting on the firm's password protected websites, bernsteinresearch.com and autonomous.com. Certain, but not all, research publications are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience.

This publication has been published and distributed in accordance with the Firm's policy for management of conflicts of interest in investment research, a copy of which is available from Bernstein Institutional Services LLC, Director of Compliance, 245 Park Avenue, New York, NY 10167. Additional disclosures and information regarding Bernstein's business are available on our website www.bernsteinresearch.com.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. This publication is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of, or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or which would subject any of the entities referenced herein or any of their subsidiaries or affiliates to any registration or licensing requirement within such jurisdiction. This publication is based upon public sources we believe to be reliable, but no representation is made by us that the publication is accurate or complete. We do not undertake to advise you of any change in the reported information or in the opinions herein. This publication was prepared and issued by entity referred to herein for distribution to eligible counterparties or professional clients. This publication is not an offer to buy or sell any security, and it does not constitute investment, legal or tax advice. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with their professional advisors in light of their specific circumstances. The value of investments may fluctuate, and investments that are denominated in foreign currencies may fluctuate in value as a result of exposure to exchange rate movements. Information about past performance of an investment is not necessarily a guide to, indicator of, or assurance of, future performance.

This report is directed to and intended only for our clients who are "eligible counterparties", "professional clients", "institutional investors" and/or "professional investors" as defined by the aforementioned regulators, and must not be redistributed to retail clients as defined by the aforementioned regulators. Retail clients who receive this report should note that the services of the entities noted herein are not available to them and should not rely on the material herein to make an investment decision. The result of such act will not hold the entities noted herein liable for any loss thus incurred as the entities noted herein are not registered/ authorised/ licensed to deal with retail clients and will not enter into any contractual agreement/arrangement with retail clients. This report is provided subject to the terms and conditions of any agreement that the clients may have entered into with the entities noted herein. All research reports are disseminated on a simultaneous basis to eligible clients through electronic publication to our client portal.

The information in this report was prepared by Bernstein solely for the internal business use of our clients. Clients may store, display, analyze, reformat and print the information in this report for this limited use only. Clients may not copy, alter, create derivative works, resell, reverse engineer, commercially exploit, share or distribute any part of the information contained herein for any purpose without Bernstein's express written consent. These restrictions include extracting data or using the content to develop indices or other products. Further, you may not use this report, or any portion of this report, to train or finetune any third-party machine learning or artificial intelligence system, or as a prompt or input into any such system. You also may not, without Bernstein's express written consent, do any of the foregoing in connection with your own internal machine learning or artificial intelligence system.

Bernstein may use artificial intelligence tools in the preparation of its materials. Any such materials are reviewed by Bernstein's research analysts prior to publication.

This report has been prepared for information purposes only and is based on current public information that we consider reliable, but the entities noted herein do not warrant or represent (express or implied) as to the sources of information or data contained herein are accurate, complete, not misleading or as to its fitness for the purpose intended even though the entities noted herein rely on reputable or trustworthy data providers, it should not be relied upon as such. Opinions expressed are the author(s)' current opinions as of the date appearing on the material only and we do not undertake to advise you of any change in the reported information or in the opinions herein.

Any references to Societe Generale herein are purely factual, based upon publicly available information, and included for comparative purposes only. They do not constitute an opinion or recommendation with respect to the securities of Societe Generale.

This publication was prepared and issued by the entity referred to herein for distribution to eligible counterparties or professional clients. The information in this report is intended for general circulation and does not constitute an offer to buy or sell any security, investment, legal or tax advice nor a personal recommendation, as defined by any of the aforementioned regulators. It does not take into account the particular investment objectives, financial situations, or needs of individual investors. The report has not been reviewed by any of the aforementioned regulators and does not represent any official recommendation from the aforementioned regulators. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with advice sought from a financial adviser regarding the suitability of the investment product, taking into account the specific investment objectives, financial situation or particular needs of any recipient of the recommendation, before the recipient makes a commitment to purchase the investment product.

The analysis contained herein is based on numerous assumptions. Different assumptions could result in materially different results. The information in this report does not constitute, or form part of, any offer to sell or issue, or any offer to purchase or subscribe for shares, or to induce engage in any other investment activity. The value of any securities or financial instruments mentioned in this report may fluctuate subject to market conditions. Information about past performance of an investment is not necessarily a guide to, indicative of, or assurance of future performance. Estimates of future performance mentioned by the research analyst in this report are based on assumptions that may not be realized due to unforeseen factors like market volatility/fluctuation. In relation to securities or financial instruments denominated in a foreign currency other than the clients' home currency, movements in exchange rates will have an effect on the value, either favorable or unfavorable. Before acting on any recommendations in this report, recipients should consider the appropriateness of investing in the subject securities or financial instruments mentioned in this report and, if necessary, seek for independent professional advice.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject the entities noted herein to any regulation or licensing requirement within such jurisdiction.

Source: Bloomberg Index Services Limited. BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively "Bloomberg"). Bloomberg or Bloomberg's licensors own all proprietary rights in the Bloomberg Indices. Neither Bloomberg nor Bloomberg's licensors approves or endorses this material, or guarantees the accuracy or completeness of any information herein, or makes any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum extent allowed by law, neither shall have any liability or responsibility for injury or damages arising in connection therewith.